

ETZIKOM AGCM, AB, Canada

WMO: 714090

Lat: 49.5528N

Lon: 111.0542W

Elev: 2772

StdP: 13.28

Time Zone: -7.00 (NAM)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-22.4	-15.1	-28.2	1.3	-22.2	-20.9	1.9	-15.0	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	29.1	90.8	61.1	87.3	60.8	83.6	59.9	65.3	80.1	63.5	79.7	61.9	78.2	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.3	86.5	70.6	58.0	79.6	68.3	56.2	74.4	66.3	31.7	80.1	30.3	79.9	29.1	78.7	73.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) N/A	N/A	N/A	-32.1	97.4	5.0	3.0	-35.7	99.5	-38.6	101.2	-41.4	102.9	-45.0	105.1		
(5)			-32.3	69.0	4.9	2.6	-35.9	70.9	-38.8	72.5	-41.5	73.9	-45.1	75.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	41.2	18.2	16.1	29.7	41.5	52.0	59.6	66.1	65.1	56.0	42.3	29.2	16.4		
	DBStd	21.25	15.20	15.76	13.96	9.05	7.91	6.05	5.46	6.45	8.66	9.57	13.92	15.15		
	HDD50	4891	987	949	630	275	72	4	0	0	36	271	627	1040		
	HDD65	8914	1452	1369	1094	704	408	183	51	79	290	705	1074	1505		
	CDD50	1669	0	0	2	21	133	292	500	470	217	31	3	0		
	CDD65	216	0	0	0	0	4	21	86	84	21	0	0	0		
	CDH74	4293	0	0	0	15	188	446	1575	1532	519	18	0	0		
	CDH80	1592	0	0	0	2	35	117	607	630	201	0	0	0		
	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	PrecAvg	14.3	0.7	0.5	0.7	1.2	1.9	3.3	1.4	1.3	1.2	0.8	0.6	0.6		
(15) Precipitation	PrecMax	20.9	1.7	1.1	2.0	2.7	4.0	8.1	4.9	3.1	2.7	2.8	1.3	1.3		
	PrecMin	7.6	0.2	0.1	0.0	0.3	0.3	0.6	0.1	0.1	0.0	0.2	0.1	0.0		
	PrecStd	3.0	0.4	0.3	0.5	0.6	0.9	1.6	1.1	0.8	0.8	0.6	0.4	0.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.3	51.2	64.4	75.5	84.2	90.4	94.6	94.5	91.6	77.0	63.9	48.9		
		MCWB	41.3	41.8	47.4	52.6	57.2	62.9	61.8	61.6	59.9	55.9	47.6	39.7		
	2%	DB	45.1	44.7	58.7	69.4	79.3	83.0	90.6	90.8	86.5	69.2	56.3	43.1		
		MCWB	37.5	37.5	45.0	50.1	55.7	60.5	62.2	60.4	58.7	51.9	44.4	36.9		
	5%	DB	41.3	40.5	53.5	63.5	74.7	79.0	86.8	87.2	80.1	63.7	51.3	39.9		
		MCWB	35.9	35.2	42.2	46.5	53.8	58.9	62.0	60.0	56.8	49.2	42.0	35.0		
	10%	DB	38.7	37.2	47.9	58.6	69.9	75.3	83.0	82.9	74.3	58.7	46.4	37.0		
		MCWB	34.4	33.2	39.1	44.1	51.8	57.4	61.2	59.5	54.5	46.2	38.8	32.9		
	0.4%	WB	41.1	42.8	48.2	53.7	58.6	66.7	69.0	65.4	62.9	56.7	49.2	41.2		
		MCDB	49.2	50.1	62.7	71.8	78.7	79.2	84.2	81.9	80.9	74.2	63.3	47.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	38.0	38.2	45.5	50.8	56.6	63.1	66.4	63.6	60.2	53.1	45.5	37.4		
		MCDB	43.8	43.8	56.8	68.0	74.9	79.4	79.9	81.4	80.5	65.9	54.6	42.5		
	5%	WB	36.2	35.5	42.8	47.5	55.0	60.6	64.5	62.2	57.7	49.9	42.4	35.3		
		MCDB	40.8	40.1	52.6	61.7	71.2	75.4	79.7	80.5	76.2	61.3	50.5	39.7		
	10%	WB	34.4	33.2	39.5	44.6	53.2	58.9	62.9	60.7	55.6	47.3	39.4	32.8		
		MCDB	38.9	37.0	47.2	57.1	67.0	72.4	77.9	78.2	73.7	57.8	45.6	37.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	18.0	19.8	20.8	24.2	25.4	24.9	29.1	29.3	28.0	24.5	20.7	18.4		
		MCDBR	18.4	23.3	28.0	34.2	34.4	32.1	35.9	37.6	38.4	32.2	26.0	20.7		
		MCWBR	13.5	17.1	16.2	17.3	14.7	14.1	13.7	13.5	15.2	16.6	15.7	15.3		
		MCDBR	17.8	22.3	27.4	32.4	29.7	29.1	29.3	30.7	35.1	30.2	24.7	20.7		
	5% WB	MCWBR	13.7	16.7	16.4	16.8	13.3	13.9	13.3	12.0	14.4	16.3	15.5	15.8		
(40) Clear-Sky Solar Irradiance	taub		0.266	0.269	0.293	0.343	0.364	0.356	0.355	0.376	0.333	0.295	0.265	0.254		
	taud		2.418	2.470	2.467	2.324	2.311	2.387	2.402	2.314	2.411	2.504	2.496	2.417		
	Ebn at Noon		255	285	294	286	284	286	285	272	274	270	255	244		
	Edh at Noon		20	24	29	37	39	36	35	37	30	23	18	17		
(44) All-Sky Solar Radiation	RadAvg		408	724	1137	1567	1854	2055	2238	1796	1275	798	452	319		
	RadStd		35	57	81	108	138	120	115	114	103	77	29	27		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (52 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page